
ML for Mathematics

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A disclaimer, and a claim

- Disclaimer: I am a mathematician: number theory, geometry, mathematical physics.
- $\text{maths} \cap \text{AI}$ becomes a frontier like $\text{maths} \cap \text{physics}$: problems and tools, both directions.
- Maths tools in ML: Riemannian manifolds in diffusion models, sheaves in graph NNs, resolution of singularities in mech interp (SLT)...
- Interpretability and formal verification are possibly the most important areas for maths in ML.
- This talk: can ML discover new mathematics?

Outline: three strands

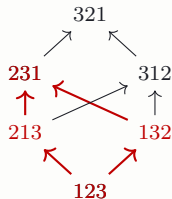
- I — learn from examples
- II — apply LLMs, "search in language space"
- III — certainty, formalization and proofs
- In each case:
 - Discuss seminal results
 - Understand when methods bite - properties of problems
 - Discuss some challenges
- recurring question: where does the machine output re-enter mathematics?
- Hang on to your 🤖, we're going to discuss some maths problems!

Strand I

Learn from examples: predict, attribute, prove

A famous example

- Permutations: $\sigma : \{1, 2, 3\} \rightarrow \{1, 2, 3\}$, denoted by $\sigma(1)\sigma(2)\sigma(3)$
- Permutation $\rightarrow$ a directed graph
- An interval $[x, y], x, y \in S_n$ subgraph between x and y
- Kazhdan–Lusztig 1979: for $x, y \in S_n \mapsto P_{x,y}(q) \in \mathbb{Z}_{\geq 0}[q]$
- combinatorial invariance (Lusztig, Dyer): isomorphic unlabelled interval graphs give the same polynomial

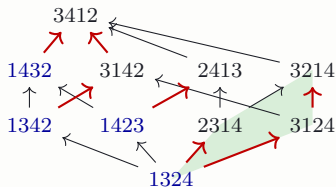


the Bruhat order on S_3 ;
in red, the interval
 $[123, 231]$.

A. Davies, P. Veličković et al., *Nature* (2021);
theorem / conj.: Blundell–Buesing–Davies–Veličković–Williamson (2021);

The experiment (Davies, Veličković,... Nature 2021)

- message-passing GNN on the bare interval graph $[x, y]$ predicts the coefficients of $P_{x,y}$
- attribution: gradients flag certain edges as overrepresented
- theorem (BBDVW): a recipe computes $P_{x,y}$ from a split of the interval into two smaller pieces, but the recipe needs the labels
- new conjecture: any valid split agrees $\implies$ combinatorial invariance. A variant led to proof for broad families (Barkley–Gaetz '25).



blue: letter 1 fixed; red: flagged edges; shaded: closing square.

Elliptic curves, and a number nobody can compute

- $E: y^2 = x^3 + Ax + B$
- Rational solutions: $E(\mathbb{Q}) \cong E(\mathbb{Q})_{\text{tors}} \oplus \mathbb{Z}^{\text{rank}(E)}$
- counting rational points is too hard, so reduce mod p and count: $a_p = (p + 1) - N_p(E)$
- represent E as a vector $(a_2, a_3, \dots)$, predict $\text{rank}(E)$
- data from an archive (LMFDB), not a generator

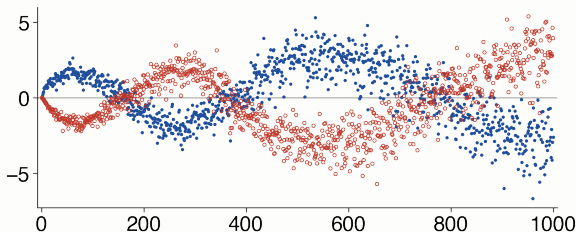
The experiment (He–Lee–Oliver–Pozdnyakov, 2022)

- logistic regression on $(a_2, a_3, \dots) \in \mathbb{Z}^{500}$, rank 0 vs 1, accuracy $> 97\%$, better than known heuristics
- experts: "you've just learned the sign in the functional equation of the associated L-function, not the rank" (follows from BSD conjecture)
- they ran another experiment but ranks 0 vs 2, accuracy: **0.996** *Why?*

Murmurations

- plotting averages over elliptic curves

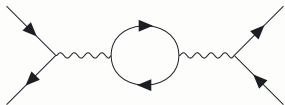
$$f_r(p) = \frac{1}{\#\mathcal{E}_r(N_1, N_2)} \sum_{E \in \mathcal{E}_r(N_1, N_2)} a_p(E)$$



- proven for certain modular forms by Zubrilina, new class of functions, then found in other classes of functions...

now: a problem I had been thinking about for a long time.

Feynman integrals: a big industry



$$\xrightarrow{P(G)}$$



- a **Feynman graph**: bookkeeping for particle interactions
- each graph $\rightarrow$ a **Feynman integral**: $P(G) = \int_{\sigma} \omega_G$
- $\omega_G = \mathcal{U}_G^a \mathcal{F}_G^b \Omega$, where $\mathcal{U}_G, \mathcal{F}_G$ are graph polynomials
- summed by loop order $\rightarrow$ the **amplitude**
- amplitudes $\rightarrow$ collider predictions, among the most precisely tested numbers in science

- Goal: compute as many Feynman integrals as possible
- Specialise to a 'toy theory': ϕ^4 , only certain graph topologies appear, no masses and momenta
- Feynman integrals carry a complexity grade: the weight, generically $2L - 3$ at L loops. On rare graphs the weight drops. Task: detect weight drop.
- A computable fingerprint: count solutions of $\Psi_G = 0$ over $\mathbb{F}_p$, like in the murmurations setup. Call the vector of these c_2 .
- Conjecture $c_2 = 0$ if and only if weight drop / simpler integral.

Feynman integrals: a big industry

- is weight drop visible in elementary graph features? 1,958 graphs up to 11 loops, 272 positives, 24 features: **AUC 0.90**
- top feature restates a theorem: 3-vertex cut $\Rightarrow$ product $\Rightarrow c_2 = 0$ (Hu–Schnetz–Shaw–Yeats). Remove those positives: **0.85**
- the 165 survivors form only 7 families, labels are constant within a family, so you learn to recognise disguised siblings (via a theorem). Hold out whole families: **0.74**
- A random test set measured recognition of previously seen results.

Feynman integrals: a big industry

- data issues: **283** irreducible families in the entire ≤ 10 -loop catalogue, exactly **7** with the drop
- the seven are conspicuous: $|\text{Aut}| \sim 20\times$ typical, and degenerate, often integer, spectra
- conjecture-shaped: exceptional weight drop lives on exceptional symmetry, but with 7 positives, a description, not a rule

Feynman integrals: a big industry

- labels are expensive: at 11 loops c_2 is known at four primes, the next prime can cost hours per graph
- why not GNNs: ϕ^4 graphs are 4-regular, so MPNN is blind to exactly the signal that works (1-WL). The fixes are the geometric-DL toolkit
- alternatively consider the “real”, decorated, Feynman graphs, but then a single $P(G)$ computation can be a paper

Feynman integrals: a big industry

- behind every invariant, one object: the graph's **Mellin motive** (Brown–Dupont–Fresán–T.), vector bundle w/ flat connection, weight drop = a fibre simplifying
- Feynman Mellin motives are representations of ‘the Cosmic Galois Group’. Can its action be learned?
- Transforming the Bootstrap: Using Transformers to Compute Scattering Amplitudes in Planar $N = 4$ Super Yang-Mills Theory (Cai et al. 2024)

Strand II

Search in language space: no dataset, an evaluator

What if there is no dataset?

- if you can score, you can search
- key idea: don't search the space of objects, search the space of programs that generate objects
- to start provide: an evaluator, and a starting program
- a program's score is the score of the object it generates, LLM proposes a handful of similar programs
- hope: the objects we care about have short descriptions which can be understood from a Python program
- history: Wagner '21 (extremal combinatorics with deep cross-entropy search), Mehrabian and others, including Adam and Petar (extremal graphs with AlphaZero), PatternBoost → FunSearch → AlphaEvolve

Flagship: 67 problems

- Georgiev, Gómez-Serrano, Tao and Wagner (Nov 2025): **67 problems**, hours each
- nearly all of the form "improve a bound on a constant"
- scorecard: 20 new, 39 matched, 8 fell short
- in author's own words: None of these results is individually important; every one of them could have been found with standard tools, the point is time (Wagner). The tool beats the literature where the literature is thin, and matches or trails it where it's deep, "long tail" (Tao).
- the expertise moved into the **evaluator**: cheap, exact, ungameable

Takeya & Nikodym over finite fields

- arena $\mathbb{F}_q^3$, q a prime power :
 - **Takeya** set - a line in every direction,
 - **Nikodym** set - a line through every point, excl. the point itself
- questions: find the smallest such sets
- previous record for Takeya in $d = 3$: $\frac{1}{4}p^3 + \frac{7}{8}p^2 + O(P)$
- AlphaEvolve record for $q = p \equiv 1 \pmod{4}$: $\frac{1}{4}p^3 + \frac{7}{8}p^2 - \frac{1}{8}$, a two-line formula over quadratic residues, proof by DeepThink
- previous record for Nikodym, by deleting random points at a fixed rate $q^3 - (2 + o(1))q^2 \log q$
- AlphaEvolve: delete surfaces, not points. As used, no better than random, but Tao repairs by hand and obtains $q^3 - (1/\log 2 + 1 + o(1))q^2 \log q$

Elementary prime counting (optional)

- $\pi(x)$ number of primes $\leq x$, define

$$C^- = \liminf_{x \rightarrow \infty} \frac{\pi(x)}{(x/\log x)} \quad C^+ = \limsup_{x \rightarrow \infty} \frac{\pi(x)}{(x/\log x)}$$

- Prime number theorem: $C^- = C^+ = 1$
- Chebyshev's game: elementary bounds on C^-, C^+ , Sylvester 1881, by hand: **0.9926**, using Möbius function
- AlphaEvolve rediscovered the Möbius function's importance, and landed at **0.938**
- Tao: "it struggled to take advantage of the number theoretic structure in the problem, even when given suitable expert hints"

A record: autoconvolution (optional)

- autoconvolution: $(f * f)(t) = \int_{\mathbb{R}} f(\tau)f(\tau - t)d\tau$

- for non-negative real f what is the best C such that

$$\|f*f\|_{L^2}^2 \leq C\|f*f\|_{L^1}\|f*f\|_{L^\infty}$$

- $C \leq 1$ by Holder, since 2010 known $0.889 \leq C$

- AlphaEvolve discovers that the gradient methods work and achieves 0.961

- numbers not important, the shape of the solution is a step-function with 50k parts

- A record, and only a record

Some remarks on the search strand

- Different possible outcomes:
 - record with no structure: autoconvolutions
 - full construction and formalized proof: Takeya $d = 3$, algebraic answer, simple enough proof
 - full construction and informal proof: Takeya $d = 4, 5$
 - seed of an idea: Nikodym, algebraic idea, had to be completed by a human
- two diagnostics: does the *object* have a short description? is the *score* an exact cheap count?

The odd zeta values

- $\zeta(s) = \sum_{n \geq 1} n^{-s}$
- even values: Euler, $\zeta(2) = \pi^2/6$; every $\zeta(2k)$ a rational multiple of π^{2k}
- all that is known about odd zeta values:
 - Apéry 1978: $\zeta(3) \notin \mathbb{Q}$
 - Ball–Rivoal 2000: infinitely many of $\zeta(3), \zeta(5), \dots$ irrational
 - Zudilin 2001: at least one of $\zeta(5), \zeta(7), \zeta(9), \zeta(11)$ irrational
- open: **is $\zeta(5)$ irrational?**

Apéry's abnormally fast fractions

- Apéry proved $\zeta(3) \notin \mathbb{Q}$ by finding fractions $\frac{p_n}{q_n}$ that approach it *abnormally fast*:

$$0 < \left| \zeta(3) - \frac{p_n}{q_n} \right| < \frac{1}{q_n^\theta}, \quad \theta = 1.08$$

- because fractions naturally repel each other, this fast convergence forces irrationality.
- the fractions came from a "miracle" recurrence relation. Where did it come from?

Beukers' geometric machine (1979)

- a family of period integrals with integer **dials** (the exponents):

$$I(a, \dots, e) = \int_0^1 \int_0^1 x^a (1-x)^b y^c (1-y)^d (1-xy)^{-e} dx dy$$

- all dials off: $I(0, 0, 0, 0, 1) = \sum_{n \geq 1} \frac{1}{n^2} = \zeta(2)$
- Beukers' choice: $I(n, n, n, n, n+1) = (-1)^n (p_n \zeta(2) - q_n)$ - Apéry's approximations
- $\zeta(3)$: the same machine, one dimension up, nine dials
- Brown recently explained how these can be viewed as **'periods of Mellin motives'**, the same objects from Strand II!

F. Beukers 1979; F. Brown, arXiv:2604.20741

The 20-Dimensional Haystack

- a candidate proof is a ray in dial space, a proof lives on a locus where the underlying family of integrals degenerates.
- **Dial counts:**
 - $\zeta(2)$: 5 dials
 - $\zeta(3)$: 9 dials
 - $\zeta(5)$: **20 dials**
- F. Brown on the 20 dials: *"hopelessly large", "significantly harder than finding a needle in a haystack"*
- scoreboard: $\zeta(3)$ needs $\theta > 1$, Apéry got **1.08** $\zeta(5)$ needs $\theta > 1$, record is **0.86**
- **the setup:** Strand II's shape if we can score the candidates.

F. Brown, arXiv:1412.6508; F. Brown, W. Zudilin, arXiv:2210.03391

Scoring a candidate: a 3-rung evaluator

- to score one point in dial space, we must compute its θ (error decay vs true size of the q_n).
- **rung 1, seconds:** error decay - numerics
 - geometric view: the differential signature
- **rung 2, minutes:** the recurrence the sequence satisfies - symbolic
 - geometric view: contiguity matrices - expensive
- **rung 3, not a computation:** θ is asymptotic, the true q_n rest on massive hidden cancellations that must be proven.
- when rung 2 fires, you don't have a theorem - you have a place to point a human proof.

Where the research goes

- in the 67 flagship problems, the evaluator was *given*: cheap, exact, ungameable
- here, search is not the bottleneck
- what is missing is a faithful, cheap score: **building the evaluator is the research.**
- calibration on the one known answer already paid off: the haystack is real, but we now know which properties of the needle actually admit a metal detector.
- the unifying target: both here and in Feynman graphs, the signal we are trying to learn is the degeneration of a Mellin motive.

Strand III

Certainty: formalization and autoformalization

What a proof assistant is

- **Lean:** write definitions, write statements the kernel checks that you can get from one to the other, build statements up until you reach the theorems you care about
- compiling a statement checks it is well-formed, not true: a statement is a type, a proof is a term of that type - the term is what you must construct

Two proofs the kernel accepts

```
theorem ex2 (a b c :  $\mathbb{N}$ ) (h1 : a = b) (h2 : b = c) : a +  
c = b + b := by  
  rw [h1]   - replace a by b everywhere  
  rw [h2]   - replace b by c everywhere
```

```
theorem ex3 (n m :  $\mathbb{N}$ ) (h : n = m) : n + 0 = m  $\wedge$  m + 0 = n  
:= by  
  constructor - one goal into two:  $\vdash n + 0 = m$  and  $\vdash m + 0 = n$   
  · rw [add_zero]; exact h  
  · rw [add_zero]; exact h.symm
```

Two proofs the kernel accepts

```
theorem ex2 (a b c : ℕ) (h1 : a = b) (h2 : b = c) : a +  
c = b + b := by  
  rw [h1]   - replace a by b everywhere  
  rw [h2]   - replace b by c everywhere
```

```
theorem ex3 (n m : ℕ) (h : n = m) : n + 0 = m ∧ m + 0 = n  
:= by  
  constructor - one goal into two:  $\vdash n + 0 = m$  and  $\vdash m + 0 = n$   
  · rw [add_zero]; exact h  
  · rw [add_zero]; exact h.symm
```

- why do this? would you read a 150-page “proof” of a major conjecture handed to you by an LLM, even as an expert?

What exists: statements, and competitions

- **mathlib**: ≈ 2.3 M lines, 9 years, roughly an undergraduate degree, jagged coverage beyond it
- statements of open problems: DeepMind's Formal Conjectures repo. Once humans agree the Lean statement captures the conjecture, checking any claimed (dis)proof is trivial

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2024	AlphaProof + AlphaGeometry	silver medal, in Lean, kernel as evaluator, one problem in minutes, up to 3 days for the others
2025	Deep Think	gold plain English, no Lean, contest hours, graded by humans
2026	AxiomProver (Axiom Math)	42/42, formal end-to-end: $\approx 8,000$ lines of Lean, ≈ 25 h of working time

Research level: the Kakeya set returns

- minimal Kakeya sets in $\mathbb{F}_p^3$: AlphaEvolve found it $\rightarrow$ Deep Think proved it $\rightarrow$ AlphaProof certified it - the full pipeline, closed (humble problem, best possible outcome)
- the paper's caveat: *only because every ingredient was already in mathlib*
- dimension 4: elliptic curves appear - beyond AlphaProof, humans checked by hand
- Nikodym sets: AlphaEvolve's construction too tangled for Deep Think to analyse, final proof entirely human (Tao '25)
- the frontier of machine certification runs out exactly where the deep objects come in

Can this scale? Two flagship formalizations

- Scholze, condensed mathematics: core theorem formalized by hand in ~ 6 months the proof came out simpler, Scholze: the check settled his own doubts about it
- Fermat's Last Theorem (Buzzard, Imperial): the Wiles–Taylor modularity theorem - a 5-year EPSRC grant (to 2029), route planned with Richard Taylor, many hands - and it covers *one branch* of arithmetic geometry
- July 2026: a needed modularity lifting theorem finished - $\approx 250k$ lines in ≈ 2 weeks *with AI tools* — now a genuine human–machine mix
- most areas of mathematics: no formalization at all

Autoformalization: sphere packing in dims 8 and 24

- the theorem: Viazovska '17 (dim 8);
Cohn–Kumar–Miller–Radchenko–Viazovska '17 (dim 24)
- humans wrote the definitions + blueprint
(Hariharan–Viazovska project, 2024), Gauss (Math, Inc.)
finished dim 8 in 5 days, dim 24 in ≈ 2 weeks, no plan given
(500k lines at peak for 8 dim, mathlib: 2.3M in 9 years)
- trust economics: read the 30-line statement, trust the kernel -
nobody referees the thousands of lines in the middle
- why here and not FLT: sphere packing's definitional tower is
shallow, FLT's is 🤯
- formalizing proofs is accelerating, definitions are the hard
currency - “compressed meaning”

Three counterexamples in nine weeks

- **20 May:** Erdős' unit distance conjecture disproved by an LLM (via Golod–Shafarevich), formalized from the axioms 5 weeks later: 1.2M lines in 3 weeks
- **11 July:** Grothendieck's 60-year-old group-scheme question: order-4 counterexample, autoformalized in 4 hours ($\sim 1,100$ lines)
- **19 July:** Jacobian conjecture (open 100 years), statement pre-formalized in Formal Conjectures, but verification trivial

Buzzard, Xena project blog, 20 July 2026

Thank you!

*“The product of mathematics is clarity and understanding.
Not theorems, by themselves.”*

— William Thurston